

Unit 5: Cloud Native Technology Part 2 – ‘Task Formative Discussion Kubernetes Cluster Deployment’

Formative Discussion Kubernetes Cluster Deployment

Step-by-Step Guide: Python 12-Factor App on Kubernetes Deployment

The Installation and Set Up on Windows guide published on the Kubernetes documentation center, was used as the source for kubectl installation. Due to the word limit of this work, this guide will focus only on Python 12-Factor App on Kubernetes Deployment.

1. Installation List:

Software	Purpose
Docker Desktop	For creating and running the container image
Minikube	For creating the local Kubernetes cluster
kubectl	For deploying and managing Kubernetes resources
Git	For managing project files

(Kubernetes, 2026)

A 12-Factor methodology Python based application, was deployed locally on Kubernetes using Docker Desktop, Minikube and kubectl on Windows 11 Pro. The 12-Factor methodology promotes portable and maintainable applications by storing configuration in environment variables and declaring dependencies explicitly (Heroku, 2024).

The required software included Docker Desktop for containerization, Minikube for creating a local Kubernetes cluster, kubectl for managing Kubernetes resources, and Git for version control, Git will be used to store the project files and maintain the portfolio; however, the use of Git is outside the scope of this guide. The installation

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instructions for kubectl followed the Install and Set Up kubectl on Windows 11 guide published by Kubernetes.

(Kubernetes, 2026)

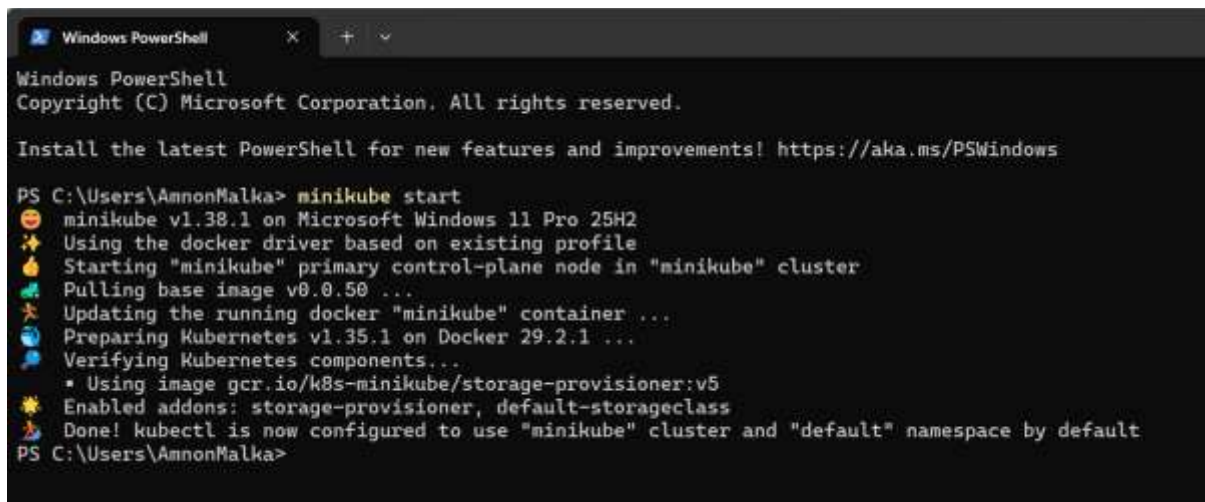
2. The Kubernetes “default” cluster was created using the following steps:

Bash....

```
minikube start
```

....

Figure 1: Kubernetes Cluster Creation



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\AmnonMalka> minikube start
🐳 minikube v1.38.1 on Microsoft Windows 11 Pro 25H2
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.50 ...
🔄 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\AmnonMalka>
```

3. The cluster status was verified using:

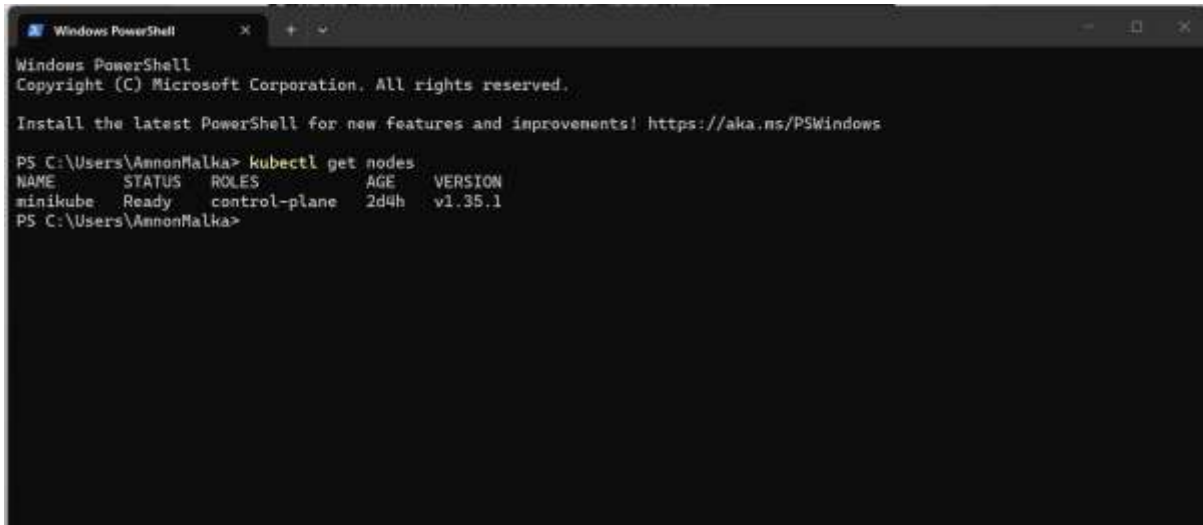
Bash....

```
kubectl get nodes
```

.....

Figure 2: Kubernetes Status Verification

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```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\AnnonMalka> kubectl get nodes
NAME      STATUS   ROLES    AGE     VERSION
minikube   Ready    control-plane  2d4h    v1.35.1
PS C:\Users\AnnonMalka>
```

The node status displayed Ready, confirming that Kubernetes was running correctly (Kubernetes SIGs, 2026).

The Python application consists of an app.py and requirements.txt, that were created before. In this deployment, the app_env environment variable was defined in deployment.yaml and injected into the application during runtime. The application had accessed this value using the os.getenv() function, allowing the same Docker image to be deployed in different environments without modifying the application source code. This demonstrates the 12-Factor principle of separating configuration from application code.

(Heroku, 2024).

Although storage of such variables demonstrates the 12-Factor approach, today, modern best practices typically use more secure secret management techniques for storing and using sensitive data, such as passwords and API keys in production. Solutions such as Kubernetes Secrets or external secret management solutions can

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help reduce the likelihood of confidential data being found in insecure location and being exploited.

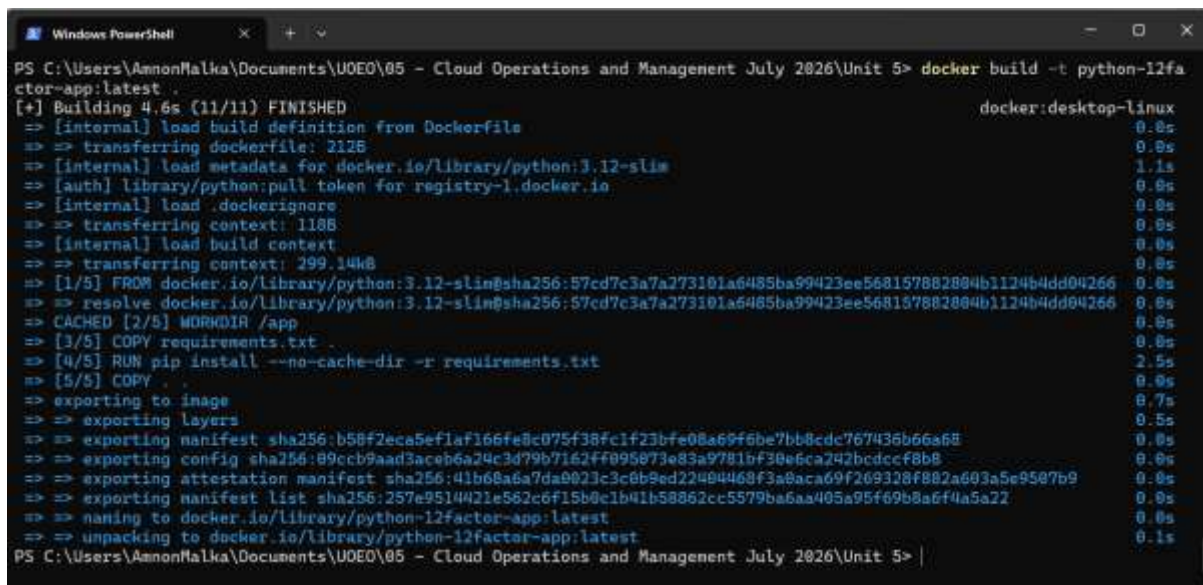
4. A Dockerfile was created to build a Docker image containing the application and its dependencies. The Docker image was built using:

Bash....

```
docker build -t python-12factor-app:latest .
```

.....

Figure 3: Docker Build Process



```
PS C:\Users\AmnonMalka\Documents\UOE0\05 - Cloud Operations and Management July 2026\Unit 5> docker build -t python-12factor-app:latest .
[+] Building 4.6s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 212B
=> [internal] load metadata for docker.io/library/python:3.12-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 118B
=> [internal] load build context
=> => transferring context: 299.14kB
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266
=> resolve docker.io/library/python:3.12-slim@sha256:57cd7c3a7a273101a6485ba99423ee568157882804b1124b4dd04266
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY
=> exporting to image
=> exporting layers
=> => exporting manifest sha256:b58f2eca5ef1af166fe8c075f38fc1f23bfe08a69f6be7bb8cdc767436b66a68
=> => exporting config sha256:89ccb9aad3aceb6a24c3d79b7162ff095073e83a9781bf30e6ca242bcdccf8b8
=> => exporting attestation manifest sha256:41b68a6a7da0023c3c0b9ed22404468f3a0aca69f269328f882a603a5e9507b9
=> => exporting manifest list sha256:257e9514421e562c6f15b0c1b41b58862cc5579ba6aa405a95f69b8a6f4a5a22
=> => naming to docker.io/library/python-12factor-app:latest
=> => unpacking to docker.io/library/python-12factor-app:latest
```

Two Kubernetes configuration files have been created. for that purpose: The deployment.yaml file which defined the application deployment. The Docker image and replica count, while service.yaml exposed the application through the Kubernetes cluster. (Kubernetes, 2025).

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5. The application was deployed using:

Bash....

```
kubectl apply -f deployment.yaml
```

```
kubectl apply -f service.yaml
```

.....

6. The deployment was verified by checking the running resources:

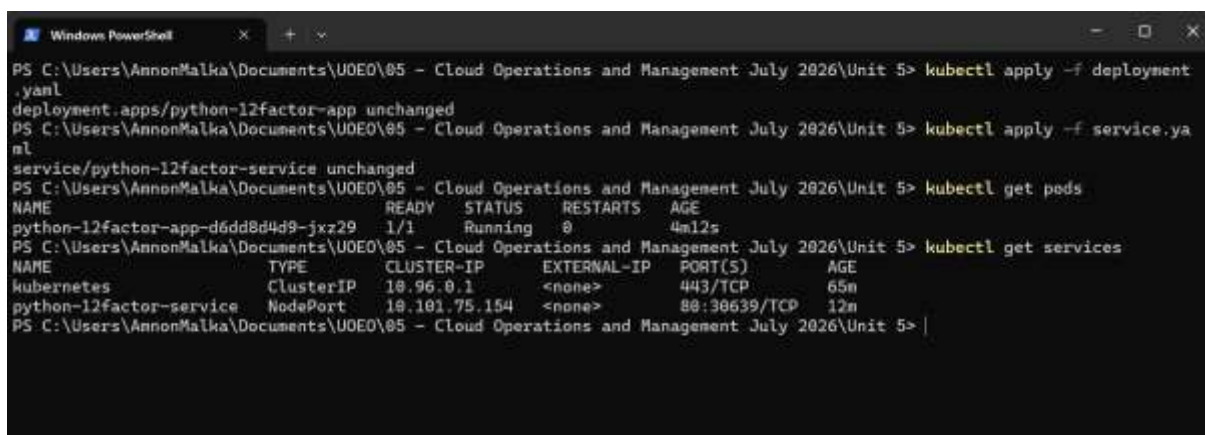
Bash....

```
kubectl get pods
```

```
kubectl get services
```

.....

Figure 4: Kubernetes Deployment and Service Verification



```
Windows PowerShell
PS C:\Users\AnnonMalka\Documents\UOE0\05 - Cloud Operations and Management July 2026\Unit 5> kubectl apply -f deployment
deployment.apps/python-12factor-app unchanged
PS C:\Users\AnnonMalka\Documents\UOE0\05 - Cloud Operations and Management July 2026\Unit 5> kubectl apply -f service.ya
ml
service/python-12factor-service unchanged
PS C:\Users\AnnonMalka\Documents\UOE0\05 - Cloud Operations and Management July 2026\Unit 5> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
python-12factor-app-d6dd8d4d9-jxz29 1/1     Running   0           4m12s
PS C:\Users\AnnonMalka\Documents\UOE0\05 - Cloud Operations and Management July 2026\Unit 5> kubectl get services
NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes                          ClusterIP   10.96.0.1     <none>         443/TCP          65m
python-12factor-service             NodePort    10.101.75.154 <none>         80:30639/TCP     12m
PS C:\Users\AnnonMalka\Documents\UOE0\05 - Cloud Operations and Management July 2026\Unit 5> |
```

The pod status displayed as running and the Kubernetes Service was successfully created. The application had been deployed successfully.

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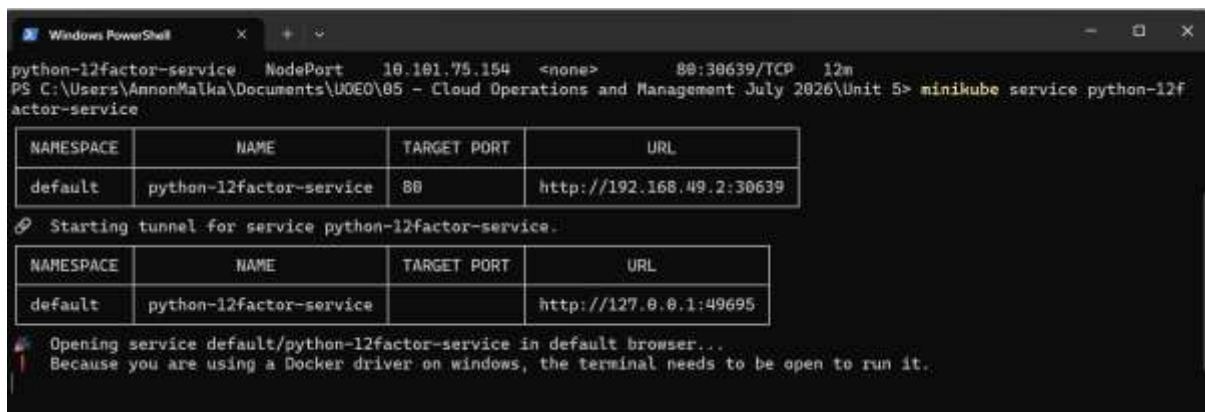
7. Finally, the application was accessed using:

Bash....

```
minikube service python-12factor-service
```

.....

Figure 5: Minikube Service Access



The screenshot shows a Windows PowerShell terminal window with the following content:

```
python-12factor-service NodePort 10.101.75.154 <none> 80:30639/TCP 12m
PS C:\Users\AnnonMalka\Documents\UOE0\05 - Cloud Operations and Management July 2026\Unit 5> minikube service python-12factor-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	python-12factor-service	80	http://192.168.49.2:30639

🔗 Starting tunnel for service python-12factor-service.

NAMESPACE	NAME	TARGET PORT	URL
default	python-12factor-service		http://127.0.0.1:49695

🌐 Opening service default/python-12factor-service in default browser...
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.

8. Opening the generated URL in a web browser confirmed that the application had been successfully deployed.

Figure 6: Application Access in Web Browser



Using this deployment it was demonstrated how to utilize Docker, Kubernetes and Minikube combined together to support the deployment of a Python 12-Factor

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principle based application. Docker was used to package the application into a portable container, Minikube provided the local Kubernetes environment, and Kubernetes was used to manage the deployment and networking through a deployment and service. The successful browser response indicated that the application was deployed correctly and that the configuration was provided through environment variables in accordance with the 12-Factor methodology.

References

- Docker Inc. (2025) Python language-specific guide. Available at: <https://docs.docker.com/guides/python/> [Accessed: 31 July 2026].
- Heroku (2024) The Twelve-Factor App. Available at: <https://12factor.net/> [Accessed: 31 July 2026].
- Kubernetes SIGs (2026) Minikube documentation. Available at: <https://minikube.sigs.k8s.io/docs/> [Accessed: 31 July 2026].
- Kubernetes (2026) Install and Set Up kubectl on Windows. Available at: <https://kubernetes.io/docs/tasks/tools/install-kubectl-windows/> [Accessed: 31 July 2026].

Use of Digital Tools

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